

A Short Guide to Lemma D.3

From least-noise Gaussian amplification to a conic stochastic bound

Abstract

Lemma D.3 turns a least-noise theorem for trace-preserving phase-insensitive displacement-covariant amplifiers into an inequality valid for trace-nonincreasing completely positive maps. The main external input is a stochastic-order result of Guță and Matsumoto. The normalization of a covariant subchannel, and the conversion of stochastic order into the displayed weighted inequality, are proved in the manuscript. This note separates those ingredients and explains each step.

1 What the lemma says informally

Among phase-insensitive amplifiers with a prescribed gain, the quantum-limited amplifier adds the least photon-number noise. Therefore, if one tests the output with a function that rewards low photon numbers, no other such amplifier can score better.

Lemma D.3 also allows a probabilistic operation, whose output can have trace less than one. Displacement covariance forces its success probability to be a state-independent constant c . After division by c , the operation is an ordinary channel to which the least-noise theorem applies; multiplication by c then gives the desired conic, or subnormalized, inequality.

2 Background

2.1 One bosonic mode and thermal states

Let $\mathcal{F}$ be the one-mode Fock space, with number basis $\{|k\rangle : k \geq 0\}$ and number operator

$$\hat{N} = \sum_{k \geq 0} k |k\rangle\langle k|.$$

For $0 < q < 1$, the thermal state is

$$\tau_q = (1 - q) \sum_{k \geq 0} q^k |k\rangle\langle k|. \tag{1}$$

Its photon number has the geometric law $t_k = (1 - q)q^k$, and its mean is $q/(1 - q)$.

The Weyl displacement operators are denoted by $D(z)$, $z \in \mathbb{C}$. A map $\Gamma : \mathcal{T}_1(\mathcal{F}) \rightarrow \mathcal{T}_1(\mathcal{F})$ has amplitude gain $\sqrt{\gamma}$ if

$$\Gamma(D(z)XD(z)^*) = D(\sqrt{\gamma}z)\Gamma(X)D(\sqrt{\gamma}z)^*. \tag{2}$$

It is phase covariant if the analogous identity holds for the rotations $e^{i\theta\hat{N}}$. “Phase insensitive” is the customary physical term for this phase symmetry.

The quantum-limited amplifier has a vacuum environment. At intensity gain $\gamma > 1$, it maps

$$\tau_q \mapsto \tau_{q^{(\gamma)}}, \quad q^{(\gamma)} = 1 - \frac{1-q}{\gamma}. \quad (3)$$

Indeed, its mean output occupation is $\gamma q / (1-q) + (\gamma-1)$, which equals $q^{(\gamma)} / (1-q^{(\gamma)})$. Notice that

$$q^{(\gamma)} - q = \frac{(\gamma-1)(1-q)}{\gamma} > 0; \quad (4)$$

amplification makes the thermal distribution broader.

2.2 First-order stochastic order

For probability laws P, Q on $\mathbb{Z}_{\geq 0}$, say that P is stochastically larger than Q if

$$\sum_k \phi(k) P_k \geq \sum_k \phi(k) Q_k$$

for every bounded increasing function ϕ . Equivalently,

$$\sum_k w(k) P_k \leq \sum_k w(k) Q_k \quad (5)$$

for every bounded nonnegative decreasing function w . Thus “larger” means that probability is shifted toward higher photon numbers. Useful decreasing tests include

$$w(k) = \mathbf{1}_{\{k \leq K\}}, \quad w(k) = e^{-sk} \quad (s > 0).$$

This is an order on photon-number distributions, not an operator order between density matrices. Phase covariance matters because the invariant output on the phase-invariant input τ_q is diagonal in the number basis, so it has an unambiguous photon-number law.

2.3 Why a covariant subchannel has constant success probability

Suppose Γ is completely positive and trace nonincreasing. There is an effect $0 \leq A \leq I$ such that

$$\mathrm{Tr} \Gamma(X) = \mathrm{Tr}(AX). \quad (6)$$

Equation (2) and cyclicity of trace imply $D(z)^* A D(z) = A$ for every z . The Weyl representation on $\mathcal{F}$ is irreducible, so its commutant consists only of scalars. Hence

$$A = cI, \quad \mathrm{Tr} \Gamma(X) = c \mathrm{Tr} X, \quad 0 \leq c \leq 1. \quad (7)$$

This is Lemma D.1 of the manuscript. In operational language, exact displacement covariance prevents the probability of success from depending on the input state.

3 The external least-noise input

The nontrivial fact imported from Guță and Matsumoto is the following:

For a trace-preserving, phase-insensitive, displacement-covariant amplifier of fixed gain, the output photon-number law on a thermal input τ_q stochastically dominates the output law of the vacuum-idler amplifier, namely the geometric law of $\tau_{q^{(\gamma)}}$.

Their characterization represents a covariant amplifier by an idler state; phase covariance makes the relevant idler number diagonal. Their least-noise lemma then identifies the vacuum idler as the stochastic minimum. The manuscript invokes Definition 2.2, Lemma 2.3, and the arbitrary-gain discussion in Section III of their paper. This classification and least-noise theorem are *not* reproved in Appendix D.

The imported statement is stronger than a comparison of mean photon number: it controls the expectation of every bounded increasing, or equivalently every bounded decreasing, test function. It is exactly this stronger form that is needed in Lemmas D.3–D.5.

4 Statement of Lemma D.3

Lemma D.3 (rephrased). *Let Γ be a completely positive trace-nonincreasing one-mode map that is displacement covariant with gain $\sqrt{\gamma}$ and phase covariant. Then, for every bounded nonnegative decreasing $w : \mathbb{Z}_{\geq 0} \rightarrow \mathbb{R}_+$,*

$$\mathrm{Tr} \left[\Gamma(\tau_q) w(\hat{N}) \right] \leq \mathrm{Tr} \Gamma(\tau_q) \mathrm{Tr} \left[\tau_{q(\gamma)} w(\hat{N}) \right]. \quad (8)$$

The right side is the mass of the possibly subnormalized output multiplied by the decreasing-test expectation under the least-noise thermal output.

5 Proof walkthrough

Step 1: isolate the trace factor

By (7), there is $c \in [0, 1]$ such that $\mathrm{Tr} \Gamma(X) = c \mathrm{Tr} X$. In particular,

$$\mathrm{Tr} \Gamma(\tau_q) = c. \quad (9)$$

If $c = 0$, then Γ is the zero map. To see this, positivity gives $\Gamma(X) \geq 0$ and $\mathrm{Tr} \Gamma(X) = 0$ for every $X \geq 0$, hence $\Gamma(X) = 0$ on the positive cone and therefore everywhere by linearity. Both sides of (8) are then zero.

Step 2: normalize the subchannel

Assume $c > 0$ and define

$$\Lambda = c^{-1} \Gamma.$$

This map remains completely positive and retains both covariance properties. Moreover,

$$\mathrm{Tr} \Lambda(X) = \mathrm{Tr} X,$$

so Λ is a trace-preserving, phase-insensitive, displacement-covariant amplifier of gain $\sqrt{\gamma}$.

Step 3: apply the imported stochastic-order theorem

Let P be the photon-number law of $\Lambda(\tau_q)$ and let

$$Q_k = (1 - q^{(\gamma)})(q^{(\gamma)})^k$$

be the law of $\tau_{q(\gamma)}$. The Guță–Matsumoto result says that P is stochastically larger than Q . Applying the equivalent decreasing-test formulation (5) gives

$$\mathrm{Tr} \left[\Lambda(\tau_q) w(\hat{N}) \right] \leq \mathrm{Tr} \left[\tau_{q(\gamma)} w(\hat{N}) \right]. \quad (10)$$

Step 4: restore the trace factor

Multiplying (10) by c and using $\Gamma = c\Lambda$ and (9) yields precisely (8). No additional Gaussian calculation is needed.

6 How to read the inequality

Taking $w = 1$ gives equality $c = c$. Taking $w(k) = \mathbf{1}_{\{k \leq K\}}$ gives

$$\Pr_{\Gamma, \text{sub}} \{N \leq K\} \leq c \Pr_{\tau_q(\gamma)} \{N \leq K\}.$$

After dividing by c when $c > 0$, the left side becomes the low-energy tail probability of $\Lambda(\tau_q)$. The inequality says that even a probabilistic covariant operation cannot conditionally produce a quieter amplified output than the vacuum-idler amplifier.

The specific weight later needed for fidelity is

$$w_k = \sqrt{\frac{(1-q)q^k}{(1-q^{(\gamma)})(q^{(\gamma)})^k}} = \sqrt{\frac{1-q}{1-q^{(\gamma)}}} \left(\frac{q}{q^{(\gamma)}}\right)^{k/2}.$$

By (4), this is bounded, nonnegative, and decreasing, so it falls exactly within Lemma D.3.

7 What is imported and what is proved

Lemma D.1 internally proves $\text{Tr } \Gamma(X) = c \text{Tr } X$ from Weyl irreducibility. Lemma D.3 proves the normalization by c and converts the decreasing-test definition into (8). The sole imported input is Guță–Matsumoto’s stochastic minimality of the vacuum-idler output.

8 Dependencies and role in the paper

The logical chain is

$$\text{Lemma D.1} + \text{Guță–Matsumoto least-noise theorem} \longrightarrow \text{Lemma D.3}.$$

Lemma D.3 is then applied one mode at a time in Lemma D.4. Lemma D.4 controls the tangent weight used in Lemma D.5, which gives the multimode fidelity upper bound in Theorem 3.4. The same tensorized inequality is also applied branchwise to the residual subinstrument in the hybrid Gaussian product bound.

The hypothesis of phase covariance is normally arranged by phase twirling. At the origin, both the thermal input and thermal target are phase invariant, and concavity of root fidelity ensures that this twirling does not decrease the payoff.

9 Minimal prerequisite checklist

To follow the internal proof, it is enough to know completely positive maps, Weyl covariance, irreducibility and commutants, and first-order stochastic dominance. The specialized ingredient is the Guță–Matsumoto least-noise theorem, which Appendix D uses as a black box.

Primary source. M. Guță and K. Matsumoto, “Optimal cloning of mixed Gaussian states,” *Physical Review A* **74**, 032305 (2006), [arXiv:quant-ph/0605161](https://arxiv.org/abs/quant-ph/0605161).